It's clear that this combination of derivatives satisfies Eq. 9.3; therefore any function of this form solves the Schrödinger equation.

Now, let's see how a function $\psi(x - ct)$ behaves. What does it look like? How does it evolve with time? Suppose we start by looking at a snapshot at $t = 0$. We can call the snapshot $\psi(x)$ because it tells us what ψ looks like at every point in space *at the specific time* $t = 0$. Of course, we don't want just any function of $(x - ct)$. We want the total probability

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx$$

to equal 1. In other words, we want $\psi(x)$ to fall off nicely to zero at infinity so that the integral doesn't blow up. Fig. 9.1 shows $\psi(x)$ schematically. With these characteristics, it makes sense to call $\psi(x)$ a wave *packet*.

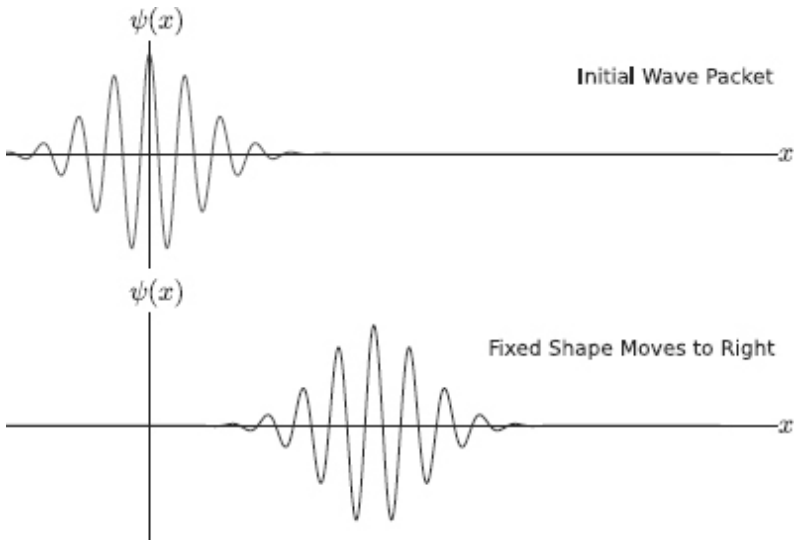


Figure 9.1: Fixed Shape Wave Packet Moving at Fixed Speed c

Now that we've described the snapshot $\psi(x)$ at $t = 0$, what happens if we let time move forward? As t increases, the wave packet keeps the exact same shape. Every feature of the complex-valued function $\psi(x, t)$ moves with uniform velocity c to the right.[1]